Simultaneous quadratic coding for ZFC consistency

Short positional certificates, a multi-output carry test, and much lower-degree nine- and ten-variable equations

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Abstract

The recent ProveIt constructions first combine a quadratic system into a quartic and then assign exponentially separated positions to its monomials. We replace both steps. A fixed 31-element positional certificate with largest position 799 distinguishes every monomial of degree at most two. Twenty-two separated target cells encode the quadratic residuals directly, and a single binary no-carry condition tests every target at once. The resulting source interface feeds the existing Jones and Sun arithmetic bridges. The principal equations have nine positive variables, degree 129,875,980 and 788 binary operations, and ten unrestricted integer variables, degree 237,526,776 and 936 operations. Their predecessors had degrees about $2.15 \cdot 10^{27}$ and $5.89 \cdot 10^{26}$, respectively. An independent improvement gives twelve positive variables at degree 941 and 121 operations. We give complete polynomial specifications, the new universal arithmetic proofs, exact finite certificates, fixed coefficient payloads, and reproducible checks. The consistency equivalence is an ordinary mathematical proof formalizable in ZFC, not a new Lean or Rocq certificate. No global minimum or new universal unknown-count theorem is claimed.

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1 Result and comparison

Throughout, $\mathbb{N} = \{0, 1, \ldots\}$ and $\mathbb{Z}_{>0} = \{1, 2, \ldots\}$. The domain is part of each assertion. A polynomial intended for positive variables must not silently be interpreted over all integers.

Theorem 1.1 (Main consistency presentations)**.** *Let F_9 and F_{10} be the fixed integer polynomials defined below and bound to the complete coefficient instance in the archive. Then ZFC proves*

$$\neg\exists \mathbf{x} \in \mathbb{Z}_{>0}^{9}\ F_9(\mathbf{x}) = 0 \iff \mathrm{Con(ZFC)},$$
$$\neg\exists \mathbf{z} \in \mathbb{Z}^{10}\ F_{10}(\mathbf{z}) = 0 \iff \mathrm{Con(ZFC)}.$$

Replacing the nine positive coordinates by nonnegative coordinates plus one gives the supplied nine-natural-variable alternative.

The theorem is an equivalence, not a proof of either side. The internal arithmetized consistency predicate is the same finite-refutation predicate used by the calibrated K215 endpoint. No unconditional two-sided independence claim follows merely from consistency, and none is made here.

1.1 What is genuinely different

There are three logically separate improvements.

1. Replace exponential Kronecker positions by a short, explicitly checked additive positional certificate.
2. Encode all quadratic residuals simultaneously instead of squaring and adding them into a quartic. The resulting source uses 136 weighted coefficient records, rather than 402 quartic records.
3. Use a direct two-block carry source. Its positivity holds before the final equation is solved, allowing substantially simpler square-root bounds in the nine-variable finalizer. The enormous dense Jones index is no longer needed.

The first step uses classical Bose–Chowla ideas; the contribution is their application through a new explicit source adapter. The second and third steps are essential to the strongest concrete reductions. Merely substituting a short list of exponents into an old theorem statement would not prove them.

Presentation	Domain	Degree	Old ops	New ops
Main nine-variable equation	$\mathbb{Z}_{>0}^{9}$	129,875,980	1500	788
Ten-variable Sun instance	$\mathbb{Z}^{10}$	237,526,776	1442	936
Low-degree twelve-variable form	$\mathbb{Z}_{>0}^{12}$	941	124	121
Same twelve-variable system	$\mathbb{Z}_{>0}^{12}$	932	120	116

For the twelve-variable comparison, the preceding lowest-degree scalar form had degree 1022 and its system had maximum degree 1013. The new system has three equations. The nine-variable predecessor in this comparison had degree $92324 \cdot 5^{32} + 296$, and the ten-integer predecessor had degree $126496 \cdot 5^{31} + 27064$.

Operations are live binary additions, subtractions and multiplications, including multiplication by numerical constants. Fixed coefficient generation, reuse of a computed value, and equality comparison in a system are free. Exponentiation is expanded into these gates; no variable exponent is allowed. This metric is not expanded monomial count, bit complexity, or coefficient height. There is no single established meaning of "simplest" that makes all these measures interchangeable.

1.2 The latest seven packages

We reviewed the corrected catalogue and packages 31–37 at commit `22e75dd6b87b` (full pin in the source manifest). The exact-source inventory records 83 Git blob checks for the initial review; additional construction-specific pins accompany their source data.

Package	Contribution retained or compared
31	Sun's ten-integer instance; a 60-operation quadratic source system; shared arithmetic in the twelve-variable norm; complete fixed tuple.
32	Fifteen- and sixteen-integer variants using a different 32-variable parent; cheaper positive norm evaluation.
33	Sixteen-integer construction and careful restrictions on cofinal square-base arguments.
34	Nine-variable low-degree finalizer, signed constructions, leading-term carrier corrections, and twelve-variable degree frontier.
35	Alternative carrier cancellations and refactored positive norms.
36	Smaller coherent compiler coefficient tuple, and rebinding of the 67-operation scalar and 59-operation systems.
37	Faster nine-variable evaluation, fourteen-integer variants, and lower-cost moderate-degree integer presentations.

The new constructions do not reduce the number of unknowns below the existing nine-positive and ten-integer counts. The old 67-operation scalar equation and 59-operation systems remain useful and are not beaten on their own operation measure. We retain package 31's complete tuple for a transparent like-for-like source comparison; we do not claim the smallest coefficient radix.

2 Fixed source, constants, and the verification boundary

The common starting point is the inherited 31-positive-variable system of 22 quadratic equations. In its 60-operation coordinate version the variables are

$$J, F, \alpha, z, f, h, i, o, j, s, w, \tau, \eta, \zeta, y, \gamma,$$
$$N, E, P, c, a, X, D, C_2, L, F_2, V, g, I, B_a, \Delta.$$

The four fixed ports are denoted β, K_0, M_C, M_F; the last is already the shifted high mask. Put

$$q = \beta J + 1, \quad k = \eta + \zeta, \quad H = 4a + 3, \quad C = q - F - \alpha,$$
$$U = L + 1, \quad Y = a - E, \quad Q = c - \eta, \quad T = P - k, \quad R = jc - V.$$

Then the equations are

$$U = q^2, \qquad N = qU$$
$$X = wq, \qquad Y = sN,$$
$$E = XY, \qquad Q = kY$$
$$T = EQ, \qquad \tau^2 = PT + 1,$$
$$D = X + ac + \gamma H, \qquad \Delta = a^2 + H$$
$$C_2 = c^2,$$
$$D^2 = \Delta C_2 + 1, \qquad k = R + 1 + hE,$$
$$R = (q - F)(N - q) - CL + (M_C + qM_F)J,$$
$$(K_0 + w)C = F + z\beta J,$$
$$F_2 = f^2, \qquad V = of - c$$
$$g = y^2 - V^2,$$
$$I = iC_2, \qquad B_a = I^2$$
$$B_a = \Delta(F_2 - 1),$$
$$y^2 - B_a g = 1.$$

Let $r_1, \ldots, r_{22}$ denote the left sides minus the right sides, in this order. Replace each of the 31 positive coordinates by $n_i + 1$ and write $q_j(\mathbf{n}) = r_j(\mathbf{n} + \mathbf{1})$. In particular, $q_5(\mathbf{0}) = 1$. The compiler and calibrated machine theorem establish

$$\exists \mathbf{n} \in \mathbb{N}^{31} \ \bigwedge_{j=1}^{22} q_j(\mathbf{n}) = 0 \quad \longleftrightarrow \quad \text{a finite ZFC refutation exists.} \tag{1}$$

The archive includes the exact sparse equations and their coordinate adapters. The older transport coordinate is $z_{\text{old}} = z + 1$; this is a genuine positive zero-set bijection, justified by the inherited compiler bounds, not by an unrestricted substitution alone.

2.1 A complete coefficient instance, not four free parameters

The active four-port instance has radix $B_0 = 2^{1064921}$. Its port bit lengths are

$$(1064921,\ 1070004,\ 1064921,\ 1064922).$$

All four actual unsigned little-endian payloads are included, with SHA-256 hashes, and must be used together. Their raw total is 533,099 bytes. The scalar source coefficients are explicit short integer polynomials in these ports; every fixed coefficient leaf is therefore specified without referring to halting or consistency. The new positional source does not use an unspecified recursively enumerable index. Its additional large leaves are explicit powers of two with fixed exponents, not exponentials in unknowns.

The mathematical dependency chain is

$$\begin{aligned} \text{new polynomial has a zero} &\longleftrightarrow \text{all shifted quadratic rows have a zero} \\ &\longleftrightarrow \text{literal K215 halts on blank tape} \\ &\longleftrightarrow \text{ZFC has a finite refutation.} \end{aligned}$$

The new source adapter and norm modifications are proved here. The native compiler, the established positive-system endpoint, and the relevant published Pell/Lucas bridge theorems remain identified dependencies. Finite circuit checks are not substitutes for these unbounded mathematical arguments. We did not compile a new end-to-end Lean or Rocq development, and we do not infer an internal ZFC theorem merely from the general soundness of either proof assistant.

3 A 31-position certificate of maximum 799

The positions, in source-variable order, are

$$\begin{aligned}(&10, 35, 55, 62, 105, 144, 145, 159, 168, 232, 245, 306, 323, 325, 353, 371,\\ &374, 382, 494, 499, 525, 566, 578, 610, 647, 681, 685, 741, 777, 783, 799).\end{aligned} \tag{2}$$

Append $w_0 = 0$. The 528 unordered pairs with repetition from these 32 positions have distinct sums modulo 1023. Thus they also have distinct sums as integers. Padding a degree-zero or degree-one monomial with zero shows that all monomials of total degree at most two have different weighted exponents $\alpha \cdot \mathbf{w}$.

This property is an exceptionally small finite certificate: enumerate $0 \le i \le j \le 31$, compute $w_i + w_j$ (mod 1023), and reject repetitions. The downstream proof needs only this explicit property, not a trusted finite-field implementation or an optimization program.

For provenance, the positions were obtained by the Bose–Chowla method in $\mathbb{F}_{32^2}$, represented by $\mathbb{F}_2[t]/(t^{10}+t^3+1)$. A primitive element is t and the subfield of 32 elements is generated by t^{33}. Taking logarithms of $t+a$ over that subfield gives a modular B_2 set. An affine unit transformation, multiplier 47 and translation -879 modulo 1023, followed by ordering yields the displayed list with zero. No optimality is claimed: the direct 528-pair verification is the correctness argument.

Lemma 3.1 (Why the classical construction works)**.** *If θ has degree two over $\mathbb{F}_q$, the logarithms of $\theta + a$, for $a \in \mathbb{F}_q$, form a B_2 set in the multiplicative cyclic group of $\mathbb{F}_{q^2}$.*

Proof. Equality of two logarithm sums gives $(\theta+a)(\theta+b) = (\theta+c)(\theta+d)$. Subtracting the two monic quadratics leaves a polynomial of degree at most one vanishing at θ. It must be zero, and the multisets of roots agree. Affine unit maps preserve equality of equal-length modular sums. □

4 The multi-output coefficient theorem

Put $W = 799$ and $h = 2W + 1 = 1599$. Reorder the residuals as

$$1, 2, 3, 4, 6, 7, \ldots, 22, 5,$$

so the final row has constant coefficient one. For the row in position j, assign target $H_j = jh$, where $1 \le j \le 22$. Hence

$$H_* = 35178, \qquad K_* = H_* + 2W = 36776.$$

Write that row as $q_j(\mathbf{n}) = \sum_{|\alpha|\le 2} p_{j,\alpha}\mathbf{n}^\alpha$ (the reordered row index is understood), and define

$$c_{j,\alpha} = (2 - |\alpha|)! \prod_i \alpha_i!\, p_{j,\alpha}, \tag{C}$$

$$D_j(t) = \sum_{|\alpha|\le 2} c_{j,\alpha} t^{H_j - \alpha\cdot\mathbf{w}}, \qquad D_*(t) = \sum_{j=1}^{22} D_j(t). \tag{3}$$

These are ordinary integer polynomials with fixed exponents. The archive's 136 records give the complete nonzero coefficient list.

Theorem 4.1 (Simultaneous extraction)**.** *For arbitrary integers $n_1, \ldots, n_{31}$, let $C(t) = (1 + \sum_i n_i t^{w_i})^2$. Then*

$$[t^{H_j}]\, C(t) D_*(t) = 2q_j(\mathbf{n}) \qquad (1 \le j \le 22).$$

If $M \geq 1$ bounds the absolute values of all $c_{j,\alpha}$, every coefficient of D_* has magnitude at most M. If $0 \leq n_i < b$, every coefficient of CD_* has magnitude at most $M(31b)^2$.

Proof. The support of D_j lies in $[H_j - 2W, H_j]$; these intervals are disjoint. The pair-sum certificate makes all positions within each interval distinct. This proves the coefficient bound for D_* without a factor of 22. A contribution from row k to target H_j requires

$$H_j - H_k = \beta \cdot \mathbf{w} - \alpha \cdot \mathbf{w}, \qquad |\alpha|, |\beta| \leq 2.$$

The right side belongs to $[-2W, 2W]$, while every nonzero difference on the left has magnitude at least $2W + 1$. Thus $j = k$. The pair-sum property then forces $\alpha = \beta$, and the multinomial coefficient cancels the factorial multiplier in (C), leaving exactly $2q_j$. For a fixed coefficient of CD_*, the triangle inequality bounds it by M times the sum of the nonnegative coefficients of C:

$$M\big(1 + \textstyle\sum_i n_i\big)^2 \leq M(31b)^2.$$

Off-target products of different rows may overlap. The last bound already accounts for them; it does not incorrectly assume all product blocks disjoint. □

Lemma 4.2 (Positivity before decoding)**.** *For every integer $B > M + 1$, $D_*(B) > 0$.*

Proof. The leading coefficient at degree H_* is two, because the final shifted row has constant one. Every other exponent is smaller. Therefore

$$D_*(B) \geq 2B^{H_*} - M\frac{B^{H_*} - 1}{B - 1} > B^{H_*} > 0.$$

□

This unconditional positivity is important for the simpler nine-variable majorants. It does not depend on the encoded equations already being true.

5 One carry test for all 22 equations

Let $b \geq 4$ and let B be a power of two satisfying

$$B \geq 4M32^2b^2.$$

Set

$$\ell = \sum_i B^{w_i}, \qquad \mathcal{T} = \sum_{j=1}^{22} B^{H_j},$$
$$N_0 = B^{W+1}, \qquad N_1 = 4B^{K_*+1}$$
$$N = N_0N_1,$$
$$M_0 = N_0 - 1 - (b-1)\ell,$$
$$T_0 = M_0 + (B-2)\mathcal{T}N_0,$$
$$J_0 = 2(1+g)^2D_*(B) + B\sum_{e=0}^{K_*} B^e,$$
$$S_0 = g + J_0N_0,$$
$$\mathcal{R} = (N-1)\big((S_0 + T_0 + 1)N + T_0 + 1\big).$$

Every sum limit and exponent is fixed. The arithmetic circuits evaluate geometric sums by division-free doubling identities.

Suppose $0 \le g < 2bB^W$. The bounds on B imply $g < N_0$ and $1 + g \le 3bB^W$. By Lemma 4.2, $J_0 > 0$; the coefficient bound for D_* gives

$$J_0 < \frac{3B}{2}\frac{B^{K_*+1}}{B-1} < N_1.$$

Also $0 \le M_0 < N_0$ and $(B-2)\mathcal{T} < N_1$. Thus $0 \le S_0, T_0 < N$. If b is a power of two, the mask M_0 has base-B digits $B-b$ in the 31 allowed cells and $B-1$ elsewhere.

Lemma 5.1 (Binary-carry identity)**.** *For $N = 2^r$ and $0 \le S, T < N$, put $X = (N-1)((S+T+1)N + T + 1)$. Then*

$$N^2 \mid \binom{2X}{X} \quad \longleftrightarrow \quad S + T \text{ has no binary carry.}$$

Proof. The exact identity

$$X = (S+T)N^2 + (N-1-S)N + (N-1-T)$$

has disjoint lower blocks, even when $S + T \ge N$. Therefore

$$\mathrm{pc}(X) = \mathrm{pc}(S+T) + 2r - \mathrm{pc}(S) - \mathrm{pc}(T) = 2r - \tau_2(S,T),$$

where τ_2 counts binary carries. Legendre's formula gives $v_2\binom{2X}{X} = \mathrm{pc}(X)$, proving the assertion. □

Theorem 5.2 (Multi-target source interface)**.** *Under the preceding hypotheses, with b a power of two,*

$$N^2 \mid \binom{2\mathcal{R}}{\mathcal{R}} \quad \longleftrightarrow \quad \exists \mathbf{n} \in [0,b)^{31} \left[g = \sum_i n_i B^{w_i} \ \wedge \ \bigwedge_j q_j(\mathbf{n}) = 0 \right].$$

Proof. Lemma 5.1 splits the no-carry test into the low block (g, M_0) and the high block $(J_0, (B-2)\mathcal{T})$. The first test decodes exactly the displayed 31 digits. This does not assume that the low-block sum has no carry into its next block: any such carry already fails the test.

For the decoded digits, put $a_e = [t^e](CD_*) + B/2$. The coefficient bound makes $0 < a_e < B$, and

$$J_0 = 2\sum_{e=0}^{K_*} a_e B^e.$$

Divide both high-block summands by two, merely shifting their binary bits. The test becomes carry-freeness of $\sum a_e B^e$ and $(B/2-1)\mathcal{T}$. At a target cell, the second summand has every low binary bit set. Because $0 < a_{H_j} < B$, carry-freeness is equivalent to $a_{H_j} = B/2$. By Theorem 4.1, this means $q_j(\mathbf{n}) = 0$. Non-target cells have zero mask and impose no extra equation. □

Every simultaneous nonnegative zero is nonzero because $q_5(0) = 1$. Choosing b as an arbitrarily large suitable power of two gives $g \ge B \ge b$ and $g < bB^W$. Thus the source interface has both soundness and the cofinal completeness needed by the arithmetic bridges. No unknown index, extra source witness, or variable exponent is introduced.

A source-transport subtlety. The optimized parent uses $z_{\mathrm{old}} = z + 1$ before its positive coordinates are shifted to nonnegative ones. This is not a bijection on positive witnesses. Soundness is immediate, but completeness uses the canonical compiler bound $z_{\mathrm{old}} \geq 2$, proved in the pinned `quartic31/QUARTIC60_PROOF.md`. That proof is an explicit dependency of both new source specializations.

6 The nine-variable equation

6.1 A fully polynomial specification

Use the mixed domain $\epsilon, \phi, n \in \mathbb{N}$ and $g, h, s, w, i, j \in \mathbb{Z}_{>0}$. Let $b = 2^{2134939} + \epsilon$ and $B = b^3$, and use the preceding source registers R, N. The 136 signed coefficient recipes have absolute values strictly below $M = 2^{2134927}$, as follows by evaluating their absolute-value bounds at the four port bit-length bounds. Thus $b \geq 4096M$. Put $G_0 = 32bB^{799}$ and define the following straight-line registers:

$$\begin{gathered}
U = N^2 w, \qquad Y = N^2 s \\
M_0 = RY, \qquad P = 2M_0^2 U, \\
K = R + 1 + h(P-1), \qquad A = M_0(U+1) \\
B_1 = 2R + 1, \qquad W_1 = bw, \\
C = B_1 + W_1 + \phi, \qquad J = 4A - 5 \\
Q = 3W_1 C - 2(W_1^2 - 1), \\
D = (A^2-1)C^2 + 1, \qquad E = iJDC^2 \\
F = (A^2-1)E^2 + 1, \\
G = A + F(CD + 1 - A), \qquad H = B_1 + jC \\
I = (G^2-1)H^2 + 1.
\end{gathered}$$

(The letters here are local registers, not the fixed source ports.) Set

$$\begin{gathered}
a_0 = (P^2-1)K^2 + 1, \qquad a_1 = DFI \\
u = JF, \\
v = QF + J(H - C), \qquad m = K^2 G_0 - 4(C - KY)^2 G_0 - 8gK^2, \\
V_0 = PK, \qquad V_1 = (AC)(AE)(GH) \\
M_d = Q + H, \\
\Omega = um, \qquad \Theta = u(n + M_d) + v - \Omega(2M_d + V_0 V_1).
\end{gathered}$$

Our equation is

$$\boxed{F_9 = (\Theta^2 - \Omega^2(a_0 + V_0^2 a_1))^2 - 4\Omega^4 V_0^2 a_0 a_1 = 0.} \tag{4}$$

All displayed powers have fixed nonnegative integer exponents; the binomial coefficient used in the proof is *not* part of this polynomial. For the nine-positive form replace ϵ, ϕ, n by the corresponding positive coordinates minus one. For the nine-nonnegative form instead replace g, h, s, w, i, j by their coordinates plus one. These are domain bijections, not permission to interpret the same equation over unrestricted integers.

6.2 The arithmetic bridge and the new finalizer

The numerical ratio theorems in the pinned `NineRatio.lean` give the following interface. For $R, N \geq 8$, $0 < b \leq R, N$, $G_0 > 0$, and the indicated witness domains, if a_0, a_1 are squares, $u \mid v$, and $m > 0$, then

$$N^2 \mid \binom{2R}{R}, \qquad b \text{ is a power of two}, \qquad 8g < G_0.$$

Conversely these conditions, with the stronger $32g < G_0$, supply the arithmetic witnesses. This is the existing *numerical* bridge, not its old exponentially spaced source specialization. The complete source and ratio proofs are included as pinned dependencies.

For the present source, positivity holds for every outer assignment: $D(B) > 0$, the low mask is positive, and $R \geq N \geq 8$, $b \leq R, N$. On the soundness range $g < 4bB^{799}$, the coefficient bound above gives $S, T < N$, so the carry theorem applies. On the completeness range the source encoding satisfies $g < bB^{799}$, which gives $32g < G_0$. This verifies both uses of the bridge without assuming a bound that is only proved afterwards.

Lemma 6.1 (Two-radical finalizer)**.** *In the displayed domain,* (4) *has a witness $n \in \mathbb{N}$ if and only if a_0, a_1 are squares, $u \mid v$, and $m > 0$.*

Proof. All a_0, a_1, u, V_0, V_1 are positive. Direct subtraction gives $V_0^2 - a_0 = K^2 - 1 > 0$. The analogous three identities for AC, AE, GH imply $V_1^2 > a_1$. Also

$$0 < v/u = Q/J + (H - C)/F < Q + H - 1 = M_d - 1.$$

If $m = 0$, then $\Theta = u(n + M_d) + v > 0$, excluding a zero. Otherwise the norm factorization gives $\Theta/\Omega = r_0 + V_0 r_1$ for real roots $r_t^2 = a_t$. If this rational sum is nonzero, squaring yields $2(\Theta/\Omega)r_0 = (\Theta/\Omega)^2 + a_0 - V_0^2 a_1$; hence both roots are rational, and then integral. A zero sum would imply $|r_1| = |r_0|/V_0 < 1$, impossible for the square root of the positive integer a_1. Thus the roots are integral and $|r_t| \leq V_t - 1$. Consequently $Z = 2M_d + V_0 V_1 + r_0 + V_0 r_1 \geq 2M_d + 1$. The vanishing factor says

$$n + M_d + v/u = mZ.$$

Its left side is positive, proving $m > 0$; integrality proves $u \mid v$. Conversely choose integral roots and set $n = mZ - M_d - v/u$. The same bounds give $n \geq 2$, and the appropriate factor is zero. □

Combining this lemma, the numerical bridge, and the simultaneous source compiler proves the nine-variable equivalence in Theorem 1.1.

6.3 Exact degree despite a leading cancellation

Let $t = \deg N = 3(H_{22} + 3W + 2) = 112731$. On the diagonal substitution assigning one indeterminate to every variable, N has leading coefficient 4 and R leading coefficient 16 at degree $3t$. The important cancellation is resolved by the exact positive identity

$$V_1^2 - a_1 = (C^2 - 1)(AE)^2(GH)^2$$
$$+D(E^2 - 1)(GH)^2 + DF(H^2 - 1).$$

The first two summands have degree $298t + 64$ and each has leading coefficient 2^{518}. Their sum has coefficient 2^{519}; the third has lower degree. Writing $E_0 = u(n + M_d) + v - 2\Omega M_d$ gives

$$\Theta^2 - \Omega^2(a_0 + V_0^2 a_1) = E_0^2 - 2E_0 \Omega V_0 V_1 + \Omega^2 V_0^2 (V_1^2 - a_1) - \Omega^2 a_0.$$

The third term uniquely dominates, with degree $576t + 6W + 140$ and coefficient 2^{1025}. Hence $\deg F_9 = 1152t + 12W + 280 = 129875980$, with diagonal leading coefficient 2^{2050}. The circuit certificate also supplies the matching multivariate upper bound. This is an exact degree, not merely a syntactic estimate or a numerical experiment.

7 Ten unrestricted integer variables

7.1 Source adaptation and equation

Use integer variables $f, g, h, k, l, w, x, y, z, m$. Here $b = 1 + 3f$, $B = \beta_s b^2$, where the exact power-of-two constant β_s is specified in `sun/MULTITARGET_INSTANCE.json`. Use the same simultaneous source registers and put $X = R$, $Y = N^2$, $C_0 = bB^{799}$. The following registers specify the unchanged Sun arithmetic tail:

$$L = lY, \qquad U = 2LX$$
$$V = 4gwY, \qquad W_0 = bw,$$
$$A = U(V + 1), \qquad B_0 = 2X + 1$$
$$C = B_0 + (A - 2)h,$$
$$P = U^2V, \qquad K = X + 1 + k(P - 2)$$
$$S_0 = 2A - 5,$$
$$T_0 = 3W_0C - 2(W_0^2 - 1), \qquad r_1 = (P^2 - 4)K^2 + 4,$$
$$\delta = 8C_0^3gK^2 - g^2[32(C - KL)^2C_0^3 + g^2K^2],$$
$$O = f^2l^2\delta, \qquad \xi = zx$$
$$D = (A^2 - 4)C^2 + 4,$$
$$E = C^2D\xi, \qquad F = 4(A^2 - 4)E^2 + 1,$$
$$G_* = 2(A + 2)(A - 2)^2E^2 - 1, \qquad \Omega_* = 16E,$$
$$H_* = C + (B_0 - C)F + \Omega_*Fy, \qquad I_* = (G_*^2 - 1)H_*^2 + 1.$$

Set

$$\Sigma = \xi S_0^5, \qquad r = T_0 + S_0y$$
$$\vartheta = (r^3 - rS_0^2 - S_0^3)(r^2 + 23S_0^2),$$
$$r_0 = b[1 + (3O - 4)b^2z^2], \qquad r_2 = DFI_*,$$
$$w_0 = 1, \qquad w_1 = r_0$$
$$w_2 = r_0(P^2 + K^2 + 3),$$
$$Z = \Sigma m - \vartheta, \qquad A_i = \Sigma^2w_i^2r_i \quad (0 \le i \le 2),$$
$$V_* = Z^2 + A_2 - A_0 - A_1, \qquad W_* = V_*^2 + 4Z^2A_2 - 4A_0A_1.$$

The ten-integer equation is

$$\boxed{F_{10} = W_*^2 - 16V_*^2Z^2A_2 = 0.} \tag{5}$$

Its exact straight-line expansion has 936 live binary operations. The multivariate degree upper bound 237526776 is attained: evaluation of the leading homogeneous circuit at the diagonal gives nonzero residues 54, 453, and 982432 modulo 101, 1009, and 1000003, respectively. One nonzero residue suffices; three provide independent replay diagnostics.

7.2 Why the new source satisfies the old tail contract

The signed tail is used as a proved arithmetic lemma, with its complete pinned proof supplied in `sun/upstream/SIGNED10_PROOF.md`, Sections 4–8. Its requirements on the source are precisely the following:

1. If b is a positive square, $f \neq 0$, and $0 \leq g < 2C_0$, the source registers have the required positive bounds. If in addition b is a power of two, the binomial divisibility condition implies a common source zero. The reverse arithmetic bridge supplies that additional condition; it does not follow from squareness alone.
2. Every source zero has arbitrarily large square choices $b = 1 + 3f$ and an encoding $b \leq g < C_0$ meeting the stronger completeness bounds.
3. $3 \mid X$ and the numerical size inequalities used by the interpolation lemma hold.

For (1), the signed positivity gadget first forces b to be a positive square; because $b = 1+3f$ and $f \neq 0$, it forces $f > 0$ and $b \geq 4$. The fixed β_s exceeds $4M32^2$, so the same coefficient and block estimates prove (1). For (2), choose arbitrarily large even-exponent powers of two congruent to 1 modulo three. For (3), $B \equiv 1 \pmod 3$ and $N \equiv 1 \pmod 3$, so the factor $N-1$ in R gives $3 \mid X$. The complete bounds, including the unsigned and signed source ranges, are proved in `sun/MULTITARGET_SOURCE_PROOF.md`.

For clarity, the tail is not a black-box appeal to an unspecified universal polynomial. Its main ingredients are the following explicit facts. The polynomial $\pi(t) = (t^3 - t - 1)(t^2 + 23)$ has a root modulo every positive integer but has no rational root. The factor S_0^5 and a valuation argument turn $\Sigma \mid \vartheta$ into $S_0 \mid T_0$. Full-residue interpolation supplies the converse witness in the required residue class. The square condition on r_0, together with $z \neq 0$, forces positivity and the square condition on b; conversely Pell saturation provides z. Finally (5) is the norm of $Z + \Sigma \sum_{i=0}^{2} \varepsilon_i w_i \sqrt{r_i}$ over all eight sign choices. The weights strictly separate the largest radical whose sign changes under an *actual* automorphism of the multiquadratic field. This forces each radicand to be a square; it does not assume three independent sign changes when the square classes are dependent. If $\Sigma = 0$, the norm is $\vartheta^8 \neq 0$, excluding the degenerate branch. The supplied proof then recovers the numerical binomial condition and the source zero. Conversely its interpolation and Pell lemmas produce every required witness.

The new result is the verified replacement of the source in this tail. The general interpolation theorem is imported from Sun's 2026 preprint, not newly reproved or machine-checked here. The older Jones-based nine-positive construction is therefore the more conservative primary recommendation.

8 A smaller low-degree twelve-positive equation

This independent improvement is useful when coefficient extraction and large degree are undesirable. It reduces the radical-separation weight rather than the number of quantified coordinates. Let all

twelve variables $J, F, \alpha, z, h, i, T, s, w, \eta, \zeta, t$ be strictly positive and set

$$q = \beta J + 1, \qquad C = q - F - \alpha$$
$$X = wq, \qquad Y = sq^3,$$
$$E = XY, \qquad k = \eta + \zeta$$
$$c = kY + \eta, \qquad a = E + Y,$$
$$H = 4a + 3, \qquad \Delta = a^2 + H$$
$$\ell = EkY,$$
$$A = 1 + \ell(\ell + k), \qquad G = 1 + \Delta c^2$$
$$b_0 = X + ac,$$
$$R_0 = ((q-1)(q-F) + \alpha)(q^2 - 1) + (M_C + qM_F)J,$$
$$N_k = k - hE - R_0, \qquad N_t = (K_0 + w)C + q - F - z(q-1),$$
$$I = ic^2, \qquad S = \Delta I$$
$$D = 1 + IS, \qquad Q = S^2$$
$$d = Q - 1,$$
$$v = cT, \qquad b_1 = c + R_0 D$$
$$P = QDv^2, \qquad U = Qb_1^2,$$
$$u = k^4 i, \qquad M = u(u + T)$$
$$L = HM + 1, \qquad f_0 = (a+2)I.$$

For the degree-941 form take

$$\sigma = b_1 + vf_0 + M((a+2)c - \ell), \qquad W = b_0 + H(t + \sigma).$$

Use the polynomial registers

$$n_0 = (HM)^2 A, \qquad d_n = dn_0$$
$$P_H = H^2 P, \qquad U_H = H^2 U,$$
$$e_H = P_H + U_H - H^2, \qquad j_0 = 4P_H U_H,$$
$$T_r = 2d(W^2 + L^2 G), \qquad K_n = [d(W^2 - L^2 G)]^2,$$
$$A_a = 2(d_n + e_H), \qquad c_0 = (d_n - e_H)^2 - j_0,$$
$$a_t = T_r - A_a, \qquad b_t = c_0 - K_n$$
$$c_1 = 16 d_n j_0,$$
$$v_r = a_t(a_t T_r + 2b_t) - c_1, \qquad u_r = b_t^2 - a_t^2 K_n,$$
$$\mathcal{N} = u_r(u_r + T_r v_r) + K_n v_r^2.$$

The single equation and its alternative three-equation system are

$$\boxed{N_k N_t(1 + \mathcal{N}) - 1 = 0}, \qquad \boxed{N_k = 1, \quad N_t = 1, \quad \mathcal{N} = 0}. \tag{6}$$

These have respectively 121 operations and degree 941, or 116 shared operations and maximum degree 932. Sharing is counted across system outputs; subtracting an imagined fixed number of gates from the scalar count gives an incorrect system count.

8.1 The uniform new separation estimate

The inherited compiler interface gives, even on the weak branch $N_k = N_t = \varepsilon \in \{1, -1\}$,

$$q \geq 16, \quad X \geq q, \quad Y \geq q^3, \quad E = XY, \quad 3q < R_0 < q^4 - 2, \quad k = R_0 + \varepsilon + hE > E.$$

It also gives $kY < c < k(Y+1)$ and $\sqrt{G} - \sqrt{A} > kY^2$, $\sqrt{A} > kY^2$. Consequently

$$Y < k/16, \qquad a + 2 < 9k/8, \qquad c < 17kY/16.$$

The definitions yield the sharper estimates

$$\frac{v\sqrt{D}}{kY^2} < \frac{uT}{8}, \qquad \frac{b_1}{kY^2} < 1 + \frac{u^2}{128}.$$

Thus

$$2(v\sqrt{D} + b_1) < MkY^2 < \min\{M\sqrt{A},\ M(\sqrt{G} - \sqrt{A})\}.$$

This directly proves the radical separation required by the inherited norm argument. It is *not* necessary for the new weight to exceed the old, much larger weight. The unchanged recovery argument forces the relevant square roots to be rational integers, while the Pell saturation argument supplies the converse witnesses. Modulo four the norm is a square, excluding the otherwise possible $\mathcal{N} = -2$ branch of the product-unit equation. For the shift, write the old witness as $t_{\text{old}} = \gamma + Mg + M\tau + y$. The existing inequalities give $M\tau > vf_0 + b_1$ and $(a+2)c - \ell < g$; hence $t = t_{\text{old}} - \sigma > 0$. This verifies completeness for the shifted equation, not just preservation of soundness.

The full norm recovery, Pell dependencies, and uniform estimate are provided in the supplied positive-variable proof and its pinned dependencies. The exact degree certificate resolves the leading cancellations via the eight radical factors. Four have degree 119; the other four have degrees 117,117,115,107. Their sum is 932, and the two outer residual factors add $7 + 2$, giving degree 941.

Shift σ	Scalar ops	Degree	System ops	Max. degree
b_1	112	961	107	952
$b_1 + vf_0$	116	949	111	940
$b_1 + vf_0 + M(ac - \ell)$	119	945	114	936
$b_1 + vf_0 + M((a+2)c - \ell)$	121	941	116	932

No row dominates all the others. The twelve-variable proposal improves the previous same-family degree/operation tradeoff but does not beat the repository's 67-operation scalar or 59-operation system when variable count and degree are ignored.

9 What is checked, what is imported, and what is not claimed

9.1 Constants are actual integers, not extra variables

The four fixed compiler ports are supplied as compressed unsigned little-endian byte strings, with raw-byte lengths and SHA-256 hashes in both instance manifests. The same coherent package-31 tuple is used throughout; its total raw size is 533099 bytes. Replacing one of these ports by a smaller number from another package would not preserve the theorem. Every coefficient recipe is a finite integer polynomial in those ports, recorded exactly, not a semantic instruction to find a ZFC proof.

All 17 live fixed leaves of each primary nine- and ten-variable DAG are additionally materialized. Each has a sign byte followed by an unsigned little-endian magnitude, compressed by gzip, and both raw and compressed hashes are given. The nine-variable leaves occupy 3464120 raw bytes and 64286 compressed bytes; the ten-variable leaves occupy 64286 compressed bytes. Duplicate values and fixed integers count as coefficients, not quantified unknowns.

The fixed Sun radix is $\beta_s = 2^{2134940}$; the Jones floor is $2^{2134939}$. These are different constants serving different source interfaces. Neither is the original fixed compiler port β. The coefficient bounds are derived from the actual 136 recipes and the four verified bit-length bounds, not from a hoped-for smaller tuple.

9.2 Operation and degree conventions

An operation is a live binary addition, subtraction, or multiplication in a straight-line DAG, including multiplication by a fixed numeral and a squaring. Fixed powers are expanded by multiplication chains. Dead gates are excluded; shared registers are counted once, including across system outputs. Literal and fixed-coefficient leaves are not arithmetic operations. Consequently operation counts do not include the offline cost of generating the fixed coefficients, nor the bit complexity of evaluating giant integers. The equations are succinct polynomial specifications, not dense monomial expansions. The exact DAG, rather than typography in this report, is the canonical machine-readable polynomial definition.

The Sun source is evaluated by homogenizing each quadratic row with B^{W-w_i} and then Horner-combining the target rows. The identity

$$\sum_j B^{H_j-2W} \sum_{|\alpha|\le 2} c_{j,\alpha} (B^W)^{2-|\alpha|} \prod_i (B^{W-w_i})^{\alpha_i} = \sum_{j,\alpha} c_{j,\alpha} B^{H_j-\alpha\cdot w}$$

proves that this is exactly the same $D(B)$, not a changed equation. It cuts a preliminary 954-operation Sun DAG to the delivered 936-operation DAG. The nine-variable evaluator uses the same homogeneous row strategy.

9.3 The proof dependency chain

The logical chain is explicit:

$$\begin{aligned} \text{finite ZFC refutation} &\iff \text{fixed compiler source zero} \\ &\iff \text{simultaneous quadratic carry witness} \\ &\iff \text{numerical Jones or Sun bridge witness} \\ &\iff F_9 = 0 \text{ or } F_{10} = 0. \end{aligned}$$

The new middle source construction and its transport conditions are proved here and in the detailed proof files. The first equivalence depends on the repository's fixed K215/compiler endpoint and coefficient binding; the arithmetic bridge depends on the identified Jones results, or on Sun's 2026 interpolation theorem for the ten-integer construction. The final norm arguments are explicit. These are ordinary mathematical proofs using arithmetical arguments formalizable in ZFC. They do not establish Con(ZFC) itself, and do not assume it in either direction.

No new complete Lean/Rocq certificate or checked object-level derivation inside a formal ZFC proof system is delivered. Existing checked theorems do not automatically certify the new coefficient specialization or source replacement. The finite verifiers below check exact polynomial identities,

source transcription, and certificates; they cannot replace the universal proof or determine whether the final equations have a zero.

9.4 Reproducible audit inventory

The archive contains the authored finite verifiers, exact JSON DAGs, source pins, manifests, fixed coefficient payloads, and inherited proof texts. The most important audits are:

- The independent quadratic checker reconstructs all 136 weighted coefficient records from the 22 original residuals; verifies all 528 pair sums; and checks 71808 cross-target incidences.
- The nine-variable checker verifies domain translations, 180 prime-field circuit/certificate replays, 15 actual-port replays, 128 finalizer completeness fixtures, 980 zero-margin exclusions, 352 independent extractions, all 17 fixed payloads, and the exact symbolic degree leader.
- The ten-variable checker reconstructs its coefficients independently, replays the homogeneous source identity, checks its payload hashes and actual port binding, and verifies nonzero leading residues at the exact degree.
- The positive-variable checks compare 240 modular radical products and 1080 explicit upper-bound cases; a separate review checks the uniform inequality argument and the source domains.
- The final comparison inventory authenticates 84 pinned source files by Git blob identifiers. The authored short-form audit records failed local simplification ideas rather than presenting them as improvements.

The final release includes a hash manifest and a driver for running the checks in an isolated temporary copy. Only newly authored scripts are run; upstream scripts, where retained for provenance, are inert source material. See `README.md` for exact commands and required Python packages. The JSON check outputs record actual results, while this report states the universal proof obligations separately. No measured build-speed improvement, global minimum, or unbounded computational search result is claimed.

10 Research directions that remain genuinely open

1. **Optimize the target layout jointly with the weights.** The equal gap $2W + 1$ is a simple sufficient condition. A finite certificate could allow selected overlapping intervals when no cross-row monomial reaches another target. Such a certificate must account for *every* source monomial and signed coefficient; random success is not a proof.
2. **Remove source registers before positional encoding.** An elimination that preserves quadracity and the canonical completeness witness might reduce both the number of positions and the number of target cells. Substitution that raises degree can erase the gain. The objective should be the resulting bridge degree, not just parent variable count.
3. **Specialize the coherent tuple only after changing the source.** The package-36 tuple may reduce coefficient storage if its complete compiler binding can be transported to this exact parent. Mixing individual constants is invalid. A separate full rebinding audit is required.
4. **Formalize the source interface once.** A generic theorem for finitely many quadratic outputs, carrying an explicit pair-sum certificate and target-isolation certificate, could replace large instance-specific positional proofs in both Lean and Rocq. The new finite checks suggest a compact certificate boundary but are not themselves that formalization.
5. **Seek a simpler arithmetic bridge.** Most remaining degree growth comes after the carry test. Removing one Pell or norm layer could matter more than further improving the positions. Any such proposal must retain unrestricted-integer soundness and exclude zero multipliers; an improvement

only on positive witnesses does not suffice for the ten-integer theorem.

The direct multi-output construction is a structural improvement, not merely a smaller coefficient or a renamed polynomial. It preserves the established variable counts while replacing exponentially large positional degrees by explicit moderate integers and reducing their live evaluation circuits.

Postscript: a newer formalization-status commit

After the construction snapshot, commit `8747904f8b973f9016434993e7ec6ae6ec7549ef` was reviewed on 23 October 2026. It adds `Computability/DiophantineConsistency/README.md` and `STATUS.md`, distinguishing mathematical equivalences, internal portability, and checked object-level ZFC derivations. It adds no new equation or checked Lean theorem. The construction pin remains unchanged. Its caution is adopted here: these new equivalences are mathematical results with explicit dependencies, not newly delivered kernel-checked derivations.

References

[1] V. Reshetnikov, *ProveIt*, pinned source `22e75dd6b87b1b199bb08c778d7cf075ae5aae08`. https://github.com/VladimirReshetnikov/ProveIt. Exact file paths and blob identifiers are in `review/SOURCE_PINS.json`.

[2] J. P. Jones, Universal Diophantine equation, *Journal of Symbolic Logic* **47** (1982), 549–571. https://doi.org/10.2307/2273588.

[3] R. C. Bose and S. Chowla, Theorems in the additive theory of numbers, *Commentarii Mathematici Helvetici* **37** (1962/63). https://repository.ias.ac.in/8578/.

[4] Z.-W. Sun, Further results on Hilbert's tenth problem, *Science China Mathematics* **64** (2021), 281–306; preprint version 7. https://arxiv.org/abs/1704.03504v7.

[5] Z.-W. Sun, *Ten unknowns for Hilbert's tenth problem over the integers*, 2026 preprint, version 2. https://arxiv.org/abs/2609.01594v2. The exact interpolation dependency is identified in the supplied signed-tail proof; no new verification of that full preprint is claimed.